

Pip

Your local macOS AI pet — sees your screen, acts on your Mac, points where to click

OpenClaw Agenthon 2026 • Team Ryann • Project Pip

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Menu bar agent

Zero paid API keys

Visible agent loop

Problem Statement

- Desktop AI assistants often live in separate chat windows — disconnected from what the user is actually doing.
- Cloud-only agents need API keys, subscriptions, and send screen context off-device — a barrier for demos and privacy.
- Most tools answer in text but do not **show** where to click, **operate** the Mac safely, or leave an auditable step log for judges and users.
- Hackathon judges need to see autonomy: capture → reason → tool → confirm → point → speak — not a black box.

Solution Overview

Pip is a menu-bar-only macOS companion: push-to-talk, screen-aware reasoning with local Ollama, visible tool steps, UI pointing via an animated pet overlay, and spoken responses — no required paid services.

- Lives in the status bar (no Dock icon); hotkey **Control + Option** for push-to-talk.
- Captures multi-monitor screenshots only during an interaction; explains and guides in natural speech.
- Runs native tools: open apps/sites, browser search, Reminders, Calendar, Notes, guarded Desktop cleanup, research PDF export.
- Fork of MIT-licensed Clicky, rebranded and extended for OpenClaw with a free local stack (Apple Speech + Ollama + macOS TTS).

AI Agent Workflow & Architecture

CompanionManager orchestrates the loop on **@MainActor**: dictation → screen capture → optional tools → Ollama chat (with images) → parse action tags → overlay + TTS.

Step	What Pip does
1	User speaks task via push-to-talk (■■■)
2	Apple Speech transcribes; ScreenCaptureKit captures screens
3	Local tools run (browser, apps, reminders) with confirmations
4	Ollama reasons over task + screenshots + tool results
5	Parse [POINT:x,y] tags → pet overlay points; macOS TTS speaks

Technical core: SwiftUI + AppKit (NSPanel overlay), ScreenCaptureKit, AVAudioEngine, CGEvent push-to-talk tap, Accessibility UI automation with URL fallback, streaming Ollama /api/chat.

Key Features, Tech Stack & Future Impact

Key features

- Pip pet overlay points to UI via [POINT:x, y, label, screenN] tags
- Visible agent step log + agent cards for task queue/status
- Wake phrase (“hey pip agent”), notch HUD, guarded Desktop cleanup & research PDF

Tech stack

SwiftUI / AppKit
Ollama (gemma3)
Apple Speech
ScreenCaptureKit
Accessibility

Future development & impact

- Deeper multi-step workflows, safer operator policies, pluggable skill packs
- Privacy-first template for students/indie devs — strong local-agent UX without cloud API spend